Chapter 3 Trigonometry

"When I trace at my pleasure the windings to and fro of the heavenly bodies,
I no longer touch the earth with my feet"

Ptolemy

3.1 Introduction


Ptolemy of Alexandaria (AD 90-168)

Trigonometry is primarily a branch of Mathematics that studies relationship involving sides and angles of triangles. The word trigonometry stems from the Greek word trigonon which means triangle and metron which means to measure. So, literally trigonometry is the study of measuring triangles. Greek mathematicians used trigonometric ratios to determine unknown distances. The Egyptians on the other hand used a primitive form of trigonometry for building Pyramids in second millinium BC(BCE). Aristarchus (310-250 BC(BCE)) used trigonometry to determine the distances of Moon and Sun.

Eratosthenes (276-195 BC(BCE)) was the first person to calculate the earth's circumference, which he did by applying a measuring system using stadia, a Ptolemy of Alexandaria (AD(CE) 90-168) standard unit of measurement during that period. The general principles of Trigonometry were formulated by the Greek astronomer Hipparchus (190-120 BC(BCE)) and he is credited as the founder of trigonometry. His ideas were used by Ptolemy of Alexandria (AD 90-168) leading to the development of Ptolemy theory of Astronomy. The most significant development of Trigonometry in ancient times was in India. Indian Mathematician and Astronomer Aryabhata (AD(CE) 476-550) defined sine, cosine, inverse cosine, inverse sine and he gave mathematical results in the form of 108 verses which included a formula for the area of a triangle.Mathematicians Brahmagupta (598 AD(CE)), Bhaskara I (600 AD(CE)) and Bhaskara II (1114 AD(CE)) are other Ancient Indians who contributed significantly to develop Trigonometry. Trigonometry was developed as a separate branch of Mathematics through the works of Johann Bernoulli (1667-1748) and Leonhard Euler (1707-1783). Euler established the fundamental results connecting trigonometric functions and complex exponential. Joseph Fourier (1768-1830) made important contribution to the study of trigonometric series. His invention of Fourier series has a wide range of applications especially in vibration analysis, electrical engineering, acoustics, optics, signal processing, image processing and quantum mechanics. In modern times, trigonometric functions are developed as mathematical functions of angular magnitudes, through the medium of which many kinds of geometrical and algebraic investigations are carried out in every branch of Mathematics and applications. Our GPS system in cars and mobile phones is based on trigonometric calculations. Advanced medical scanning procedures such as CT and MRI, used in detecting tumors, involve sine and cosine functions.